Intraoperative Sonographically Assisted Radioactive Iodine 125 Plaque Brachytherapy for Choroidal Melanoma

Visual Acuity Outcome

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Objectives—The purpose of this study was to present a retrospective series of cases from a single Canadian academic center assessing visual acuity outcomes after intraoperative sonographically assisted iodine 125 (^{125}I) plaque brachytherapy treatment.

Methods—The cases of 28 patients (16 male and 12 female; mean age ± SD diagnosis, 62.3 ± 15 years) with choroidal melanoma treated with ^{125}I plaque brachytherapy using intraoperative sonography between 1997 and 2002 were reviewed.

Results—The mean longitudinal, transverse, and depth dimensions were 11.4, 10.6, and 4.7, respectively. The median follow-up was 48 months (range, 3–102 months) for our cohort of patients. The prescribed dose was 85 Gy to a height of 5 mm (for an apex height ≤5 mm) or to the tumor apex (for an apex height >5 mm). Five years after ^{125}I plaque brachytherapy, all tumors had regressed in their longitudinal, transverse, and depth dimensions. The prebrachytherapy tumor depth ($P = .023$) and sclera dose ($P = .036$) were found to significantly affect visual acuity after plaque brachytherapy at 24 months. One recurrence was recorded 6 years after plaque brachytherapy.

Conclusions—This study supports ^{125}I plaque brachytherapy as an efficacious treatment for patients with choroidal melanoma, and intraoperative sonography may help with optimizing tumor control. In addition, to our knowledge, this study is the first to report the sclera dose as a significant predictor of visual acuity.

Key Words—brachytherapy; choroidal melanoma; iodine 125; sonography

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Abbreviations

COMS, Collaborative Ocular Melanoma Study; logMAR, logarithm of the minimum angle of resolution; ^{125}I, iodine 125

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Choroidal melanoma is the most common intraocular ocular tumor and is treated by enucleation or by an eye-preserving technique, such as external beam radiotherapy, plaque brachytherapy, or thermotherapy.[1,2] There are also studies investigating the application of focused ultrasound for targeted drug delivery.[3] Since the Collaborative Ocular Melanoma Study (COMS) showed no difference in survival between enucleation and plaque brachytherapy, iodine 125 (^{125}I) plaque brachytherapy continues to be a popular eye-preserving treatment.[4] Plaque brachytherapy is a treatment that offers the patient the option of keeping the eye while possibly maintaining reasonable vision in the affected eye as well.[5,6]

Although ^{125}I is the most popular plaque brachytherapy technique, local treatment failures have been reported to occur in 10% to 20% of cases.[4,7,8] Optimal placement of the ^{125}I plaque helps decrease such morbidity.

Intraoperative sonography has become an increasingly important technique for appropriately localizing plaque placement to the choroidal melanoma. Other centers with shorter follow-up periods have reported satisfactory results with intraoperative sonography, stating that it aids in optimizing the position and hence may help ensure optimal treatment and outcomes.[9–11]

In this article, we present the long-term visual acuity outcomes of patients with choroidal melanoma who were treated with ^{125}I plaque brachytherapy with intraoperative sonographic localization and confirmation of the tumor-plaque relationship at a single Canadian academic center.

Materials and Methods

Clinical records and sonographic localization data were analyzed for all patients treated with ^{125}I plaque brachytherapy at the McGill University Health Center between January 1997 and December 2002. The review was done after obtaining written consent from the local institution according to internal ethics guidelines for retrospective reviews. A total of 28 patients were included in this analysis (16 male and 12 female; mean age ± SD at diagnosis of 62.3 ± 15 years). Table 1 shows tumor characteristics, and Table 2 shows treatment characteristics.

The diagnosis was determined on the basis of clinical and radiographic criteria, including tumor location via a slit lamp, mean basal tumor diameter, and tumor thickness (determined by A- and B-scan sonography). All patients had a medical history and a complete physical examination done at the first consultation visit. Before initiating treatment, patients had a metastatic workup that included blood tests, chest radiography, and abdominal sonography.

All eyes were assessed preoperatively and postoperatively for the best corrected distance visual acuity using a Snellen chart. The logarithm of the minimum angle of resolution (logMAR) was used for analysis and reporting of visual acuity.

The COMS protocol was followed for categorization of tumor size12 as follows: small tumor, depth of 2.5 mm or less, base (transverse) diameter of less than 10.0 mm, or both; medium tumor, depth of 2.6 to 10.0 mm, base diameter of less than 16.0 mm, or both; and large tumor, depth of greater than 10.0 mm, base diameter of 16.1 mm or greater, or both. The tumor location was also assessed. Tumors were considered equatorial if 50% or more of the diameter was located within the equator area, posterior to the equator if 50% or more of the tumor diameter was posterior to the equator, anterior if more than 50% of the tumor diameter was anterior to the equator, and close to the optic nerve if it was centered within 5° from the optic nerve.

All tumors were treated with an ^{125}I plaque. The plaque was inserted by an ophthalmic surgeon at the tumor site, and the position confirmed by intraoperative sonography (standardized A-scan probe, 8 MHz, parallel beam; standardized B-scan probe, 10 MHz; I^{3}System-ABD; Innovative Imaging, Inc, Sacramento, CA).[10] A dummy nonradioactive plaque was first placed under sonographic localization to set the tumor-plaque relationship. Once a satisfactory tumor-dummy plaque relationship was established, the dummy plaque was removed, leaving the sutures in place. Next, the ^{125}I plaque was placed in the predetermined location with the sonographic tumor-plaque position reconfirmed at the finish. Adjustments, if necessary, were made according to the sonographic confirmation findings. The prescription followed the revised COMS guidelines,[12] and the calibration certificates for all plaques were traceable to the National Institute of Standards and Technology National Standards Laboratory (Gaithersburg, MD). The design of the plaque allows accurate calculation of the dose to specific structures, such as the sclera and tumor apex. According to the COMS

Table 1. Preoperative Tumor Characteristics (n = 28)

Characteristic	Value
Longitudinal dimension, mm	11.4 ± 2.6
Transverse dimension, mm	10.6 ± 2.1
Depth, mm	4.7 ± 1.9
Left eye, n (%)	21 (75.0)

Data are presented as mean ± SD where applicable.

Table 2. Plaque Brachytherapy Characteristics (n = 28)

Characteristic	Value
Treatment time, h	99.2 (77.0–123.0)
Prescription point dose, Gy[a]	84.62 (77.9–94.0)
Tumor apex dose, Gy[b]	98.4 (80.9–155.5)
Internal sclera dose, Gy[c]	286.3 (226.7–545.0)

Data are presented as median (range).
[a]The prescription point dose is the dose delivered to 5 mm if the tumor height is less than 5 mm or to the tumor apex if the tumor height is greater than 5 mm, at a point perpendicular to the central axis of the plaque.
[b]The tumor apex dose is the dose delivered to a point at the maximum tumor height, calculated along a plane perpendicular to the central axis of the plaque.
[c]The internal sclera dose is the dose delivered to the sclera, calculated along a plane perpendicular to the central axis of the plaque.

guidelines, the plaque was delivered a dose of 0.5 to 1.25 Gy/h for a total dose of 85 Gy to a height of 5 mm (for an apex height ≤5 mm) or to the tumor apex (for an apex height >5 mm). The treatment time lasted a number of days depending on the initial dose rate. The internal sclera dose was calculated along the central axis of the plaque. The ophthalmic surgeon removed the plaque at completion.

Follow-up visits occurred 3 and 6 months after brachytherapy biannually for the following 5 years and annually thereafter. At follow-up visits, visual acuity, intraocular pressure, slit lamp, and dilated fundal examinations were performed for both eyes, in addition to sonographic examinations for the involved eye. Furthermore, a complete systemic assessment was also performed, which included chest radiography and liver function tests.

Descriptive statistics were used to describe patient, tumor, and treatment characteristics as well as outcomes. Outcomes were evaluated by using the raw data at each follow-up point and the change from baseline. The Wilcoxon signed rank test was used to investigate changes in tumor dimensions at different time points. Linear regression was used to explore selected variables as potentially prognostic for changes in the logMAR scale from baseline to 12 and 24 months. Logistic regression was used to explore associations with potential adverse effects (enucleation, recurrence, metastasis, radiation retinopathy, and cataract) at any time during follow-up. A binary outcome rather than a time-to-event outcome was used because of the small numbers and inconsistent timing capture. Variables considered as potentially prognostic included age, sex, lesion size, lesion location, treatment time, total radiation dose to the prescription point, radiation dose to the apex, radiation dose to the internal sclera, and treatment time. No interpolation for missing data was performed. Analyses were performed with SAS version 9.1 software (SAS Institute Inc, Cary, NC), and all tests were 2 sided with $P \leq .05$ considered statistically significant.

Results

The median follow-up for our cohort of patients was 48 months (range, 3–102 months). At the time of diagnosis, 20 patients had visual acuity of 0.54 or lower. For the 15 patients assessed 24 months after plaque brachytherapy, the mean deterioration in visual acuity ± SD was 0.3 ± 0.4. For 10 patients, the mean deterioration was 0.6 ± 0.7 at the 48-month follow-up.

Five years after ^{125}I plaque brachytherapy treatment, all tumors had regressed in their longitudinal, transverse, and depth dimensions. One patient had a tumor recurrence 6 years after treatment. One year after reirradiation with external beam radiotherapy, this patient maintained visual acuity of 2.0. One patient had metastatic melanoma 18 months after brachytherapy. Seven patients had cataracts, 5 had radiation retinopathy, and 4 had retinal detachment after brachytherapy. Two patients required enucleation: one 6 months after treatment and the second 24 months after treatment.

Regression analysis revealed that both the tumor depth before plaque brachytherapy and sclera dose were statistically significant factors involved in decreased visual acuity at 24 months. None of the tumor or treatment characteristics were significantly predictive of enucleation, recurrence, metastasis, or radiation retinopathy (Table 3).

There was a statistically significant decrease in the 3 tumor dimensions at 12, 24, and 36 months compared with the baseline presentation at diagnosis (Figures 1–3). At 12 months, the median decreases in the longitudinal dimension, transverse dimension, and depth were 1.75, 1.95, and 0.80 mm, respectively ($P < .001$ for all). At 24 months, the median decreases in the longitudinal dimension, transverse dimension, and depth were 2.10, 1.75, and 1.15 mm ($P < .001$ for all). At 36 months, the median decreases in the longitudinal dimension, transverse dimension, and depth were 3.05, 1.875, and 1.125 mm ($P < .001$ for all).

Figure 4 shows B- and A-scan images from a 67-year-old man with left choroidal melanoma. The patient received ^{125}I plaque brachytherapy in 1999 when the tumor measured 14.0 × 11.25 × 3.75 mm (longitudinal × transverse × depth) and the patient had visual acuity of 0.8 (Snellen 20/25). The preoperative A-scan image (Figure 4B) showed

Table 3. Analysis of Factors Prognostic for Changes in the LogMAR Scale

Factor	*P*
At 12 mo	
Preoperative longitudinal dimension	.24
Preoperative transverse dimension	.17
Preoperative depth	.11
Location	.087
Prescription point dose	.44
Apex dose	.89
Sclera dose	.24
At 24 mo	
Preoperative longitudinal dimension	.39
Preoperative transverse dimension	.34
Preoperative depth	.023
Location	.060
Prescription point dose	.58
Apex dose	.51
Sclera dose	.036

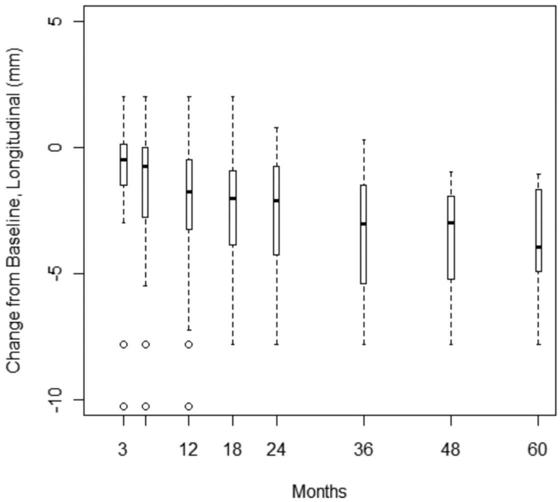


Figure 1. Baseline longitudinal tumor dimensions versus time after ^{125}I plaque brachytherapy.

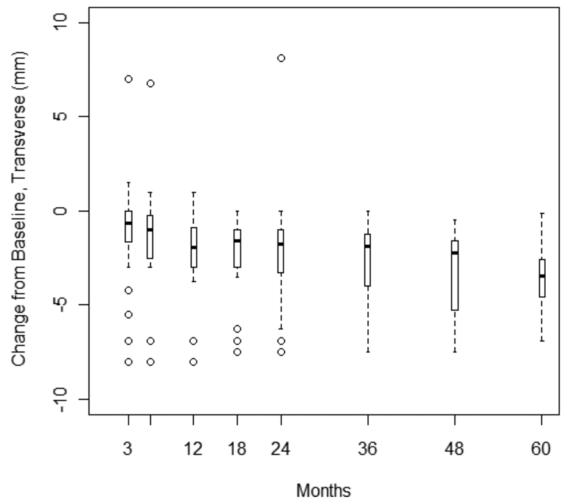


Figure 2. Baseline transverse tumor dimensions versus time after ^{125}I plaque brachytherapy.

medium-high internal reflectivity. Sonography confirmed the nonradioactive plaque position (Figure 4C) and the ^{125}I plaque position (Figure 4D) intraoperatively. At follow-up, the A-scan image (Figure 4E) showed continued medium-high internal reflectivity, and the B-scan image (Figure 4F) showed reduced dimensions of 10 × 10 × 2.7 mm, with visual acuity maintained at 0.8 (Snellen 20/25).

Discussion

In this retrospective cohort study of patients with choroidal melanoma treated with ^{125}I brachytherapy, we found that the prebrachytherapy tumor depth and sclera dose had a significant effect on visual acuity. The significance of the tumor depth as an effect on the visual acuity outcome has been supported by other studies.[13] In COMS report 16, visual acuity was assessed 3 years after ^{125}I plaque brachytherapy, and the tumor depth at presentation was found to be a significant factor associated with decreased visual acuity. To our knowledge, the sclera dose as a significant contributor to visual acuity outcomes has never been reported previously.

As described by Radin et al,[14] scleral necrosis is a rare complication after plaque brachytherapy. Although our plaques were delivered an approximate median dose of 60 Gy more to the internal sclera than what was reported by Radin et al,[14] we had no reports of scleral necrosis.

Several single-institution reports and the COMS have described decreased visual acuity after brachytherapy, although the magnitude of changes in visual acuity varied between studies. Shields et al[15] showed that patients

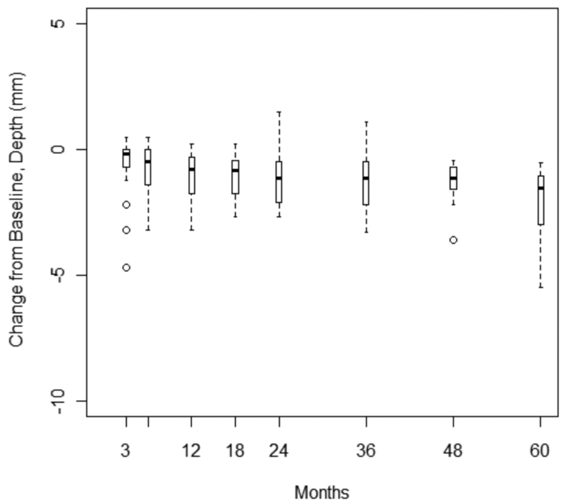
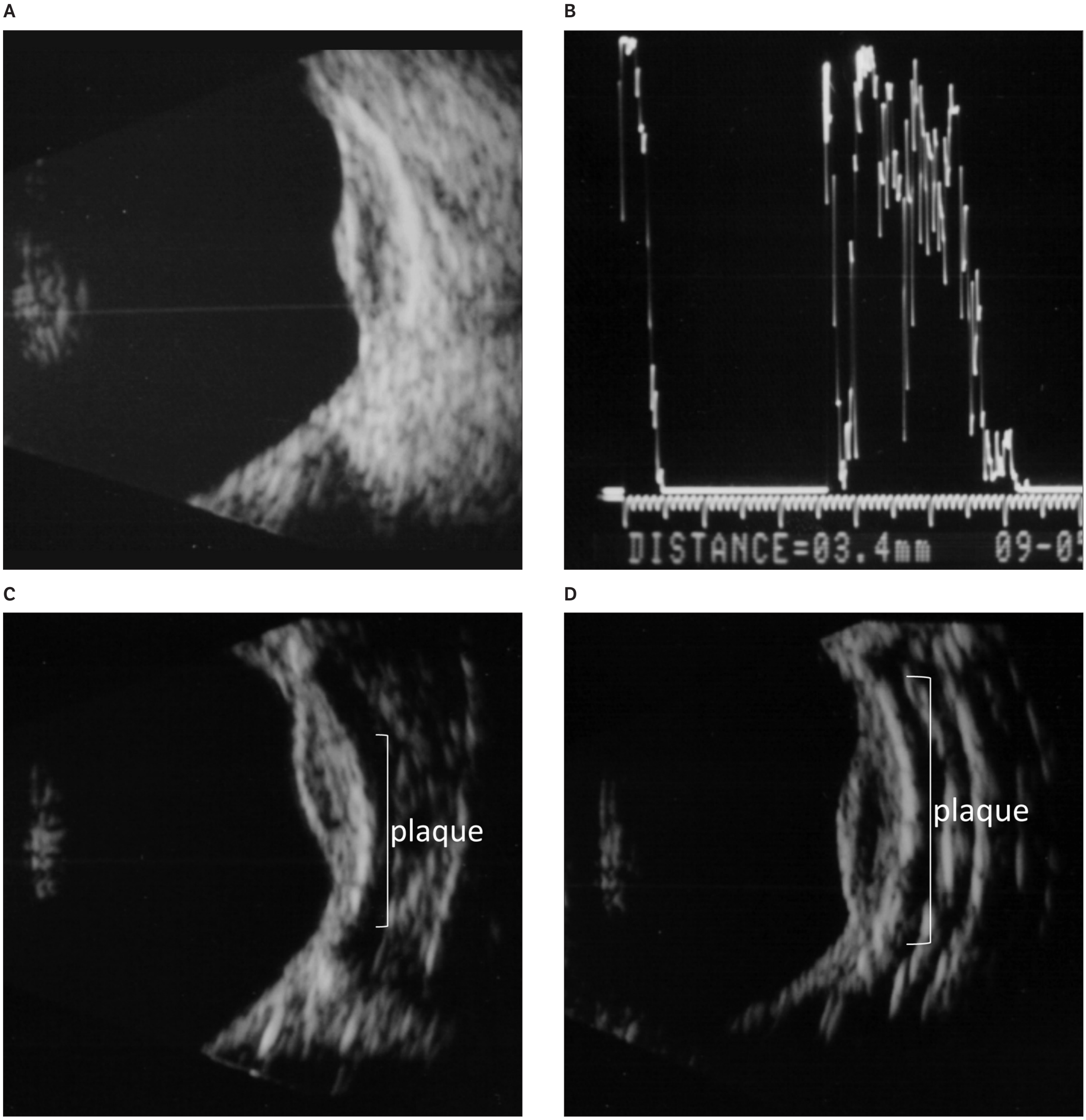


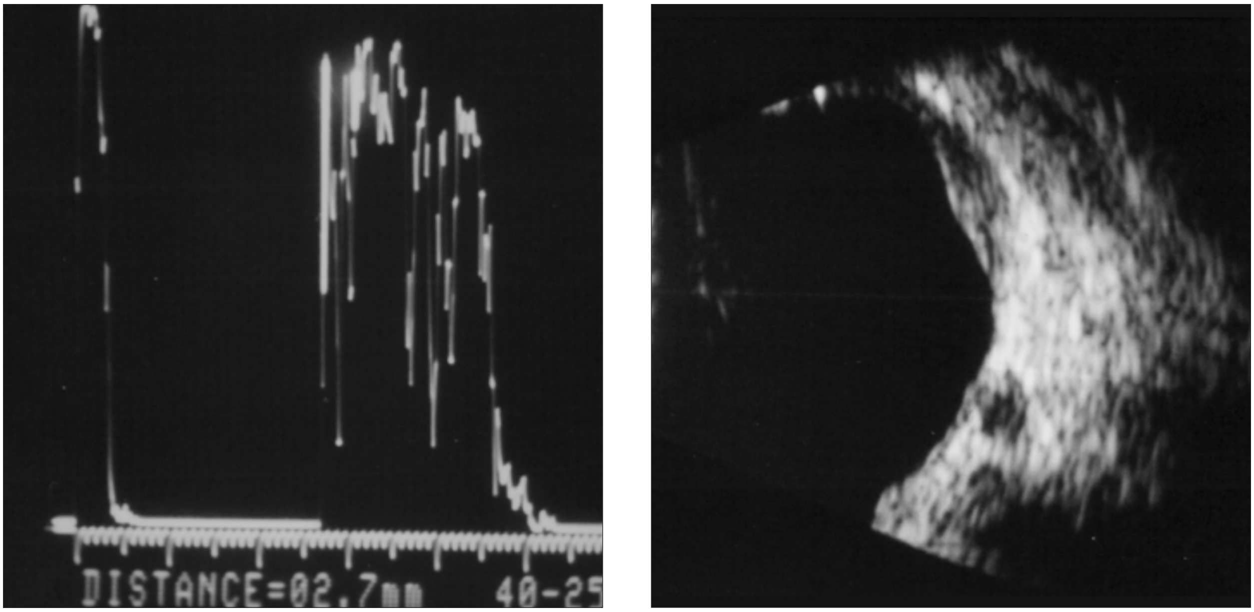
Figure 3. Baseline tumor depths versus time after ^{125}I plaque brachytherapy.

with good vision (Snellen ≤20/100) before plaque brachytherapy ultimately had poor vision acuity of 20/200 (logMAR = 1.0) or worse in 13%, 57%, 89%, and 97% of cases at 1, 5, 10, and 15 years, respectively, after treatment. The median 36-month visual acuity reported by the COMS was 0.8.[13] Our center maintained reasonable visual acuity of 0.3 and 0.3 (range, 0.2–2) at 24 and 36 months after brachytherapy, respectively, in 15 patients with available data.

The choroidal melanoma tumors in our cohort had an excellent response, with all tumors regressing through 5 years of follow-up. Mean tumor measurements at diagnosis of 11.1 × 10.2 × 4.4 mm (longitudinal, transverse, and depth, respectively) in 28 patients had decreased to 7.4 × 7.4 × 2.9 mm at 36 months after treatment in 18 patients. Unfortunately, 1 patient did have a recurrence and went on to receive external beam radiation. In a retrospective series of 63 patients with choroidal melanoma, Jones et al[16] achieved strong local tumor control in 86.9% and also found that the macular dose rate was significant for visual acuity loss 36 months after brachytherapy.

Figure 4. Sonography at different stages from a 67-year-old man with left choroidal melanoma. **A**, Intraoperative sonogram of choroidal melanoma. **B**, A-scan image with medium-high reflectivity before treatment. **C**, The nonradioactive plaque at the tumor required readjustment of the posterior margin. **D**, The plaque was adjusted with good coverage of borders (continued).





E F

Figure 4. (continued) **E**, A-scan image showing persistent medium-high reflectivity after plaque brachytherapy. **F**, Melanoma regression 2 years after treatment.

Current practice using sonographically assisted plaque placement optimizes radiation to the primary lesion and local control while minimizing side effects in normal ocular structures. Using intraoperative sonography, Chang et al[11] described repositioning of plaques 36% of the time and reported no local or distant recurrences 2 years after treatment, attributing these findings to optimal sonographic plaque placement. Their experience is in contrast to that of the COMS, which used intraoperative sonographic placement in only 80% of their procedures and reported a risk of treatment failures of 10.3%.[4] Al Mahmoud et al[17] described McGill University Health Center's experience with using a dummy plaque under sonographic guidance. Ninety percent of the patients (including the cohort in this study) had the dummy plaque readjusted for an optimal plaque-tumor relationship. Our results appear to indicate satisfactory outcomes by using intraoperative sonography for choroidal melanoma ^{125}I plaque brachytherapy placement, with only 1 local treatment failure 6 years after treatment.

In conclusion, although acknowledging that our study had a small number of patients, our center's long-term follow-up experience lends additional support for ^{125}I plaque brachytherapy as an efficacious modality for choroidal melanomas. In addition, our center's experience with intraoperative sonographic placement provides long-term evidence that it is of benefit for obtaining good local and distant control and in maintaining some visual acuity.

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